

COGS 9 | Week 1

The background is a dark blue gradient. On the right side, there is a vertical column of binary code (0s and 1s) in a light blue color. A yellow sine wave curves across the middle of the image. In the foreground, there is a candlestick chart with red and green bars, overlaid with a yellow trend line. The chart appears to be a financial or data visualization.

Data Science Questions
&
Ethics & privacy

Brendan Tomoschuk

Upcoming + Reminders

Lecture slides: Now available on the course website within 2-3 days after lecture



Read: Reading 1, *50 Years of Data Science*

All due by 11:59pm PT



Assignments (find links on course website):

- Pre-course survey (**DUE: Wed - Week 2**)
- Syllabus quiz (**DUE: Wed - Week 2**)
- Reading quiz 1 (**DUE: Wed - Week 2**)

Section begins **friday!**

Discussion sections are for:

-  Discussing course material & readings in more depth
-  Meeting your classmates & group project mates
-  Working on the group project
-  Getting help from & getting to know your TA/PLA!

You may go to a different section than the one you are enrolled in

You **DO NOT** need to ask us for permission!

Webclicker will log your participation regardless of which section

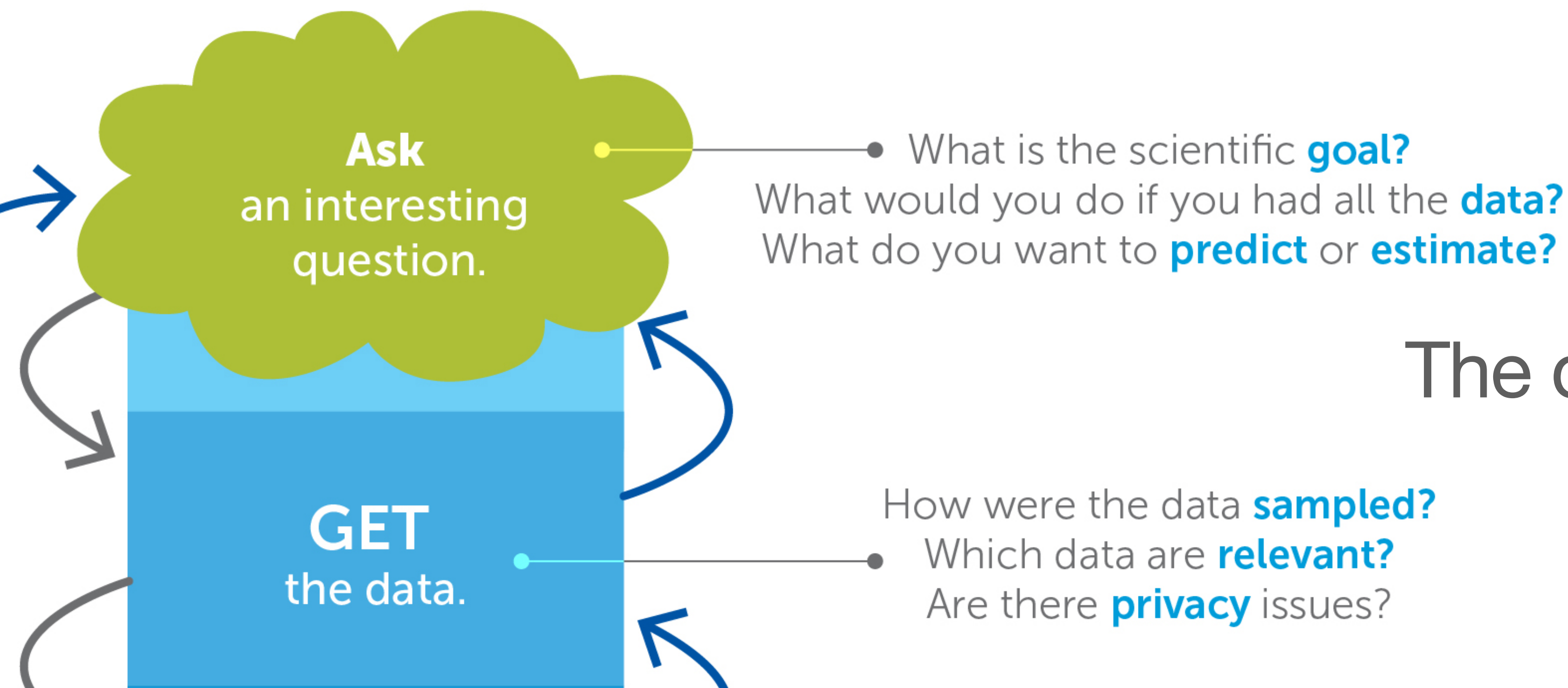
Group project

Groups of 4-5 students:

1. Form your own group
 - Preferably in the same section, but if not: everyone needs to commit to meeting outside of section
2. Be matched to a group based on interests / working style
 - [Group matching form](#) released later this week!

Last time: data science questions

- The importance of **starting with a question**
- Turning a general question into a *specific, measurable* one
- Different types of **data science goals**: describe, predict, cause-and-effect, or decide / optimize



The data science process

Specific, measurable questions

Do Tiktok videos with dogs go more viral?

- Let's make it better!
- Do tiktok viewers between the ages of 18 and 20 view more videos with a dog in at least 50% of the frames than videos with cats in 50% of frames within the first 10 days of being posted?

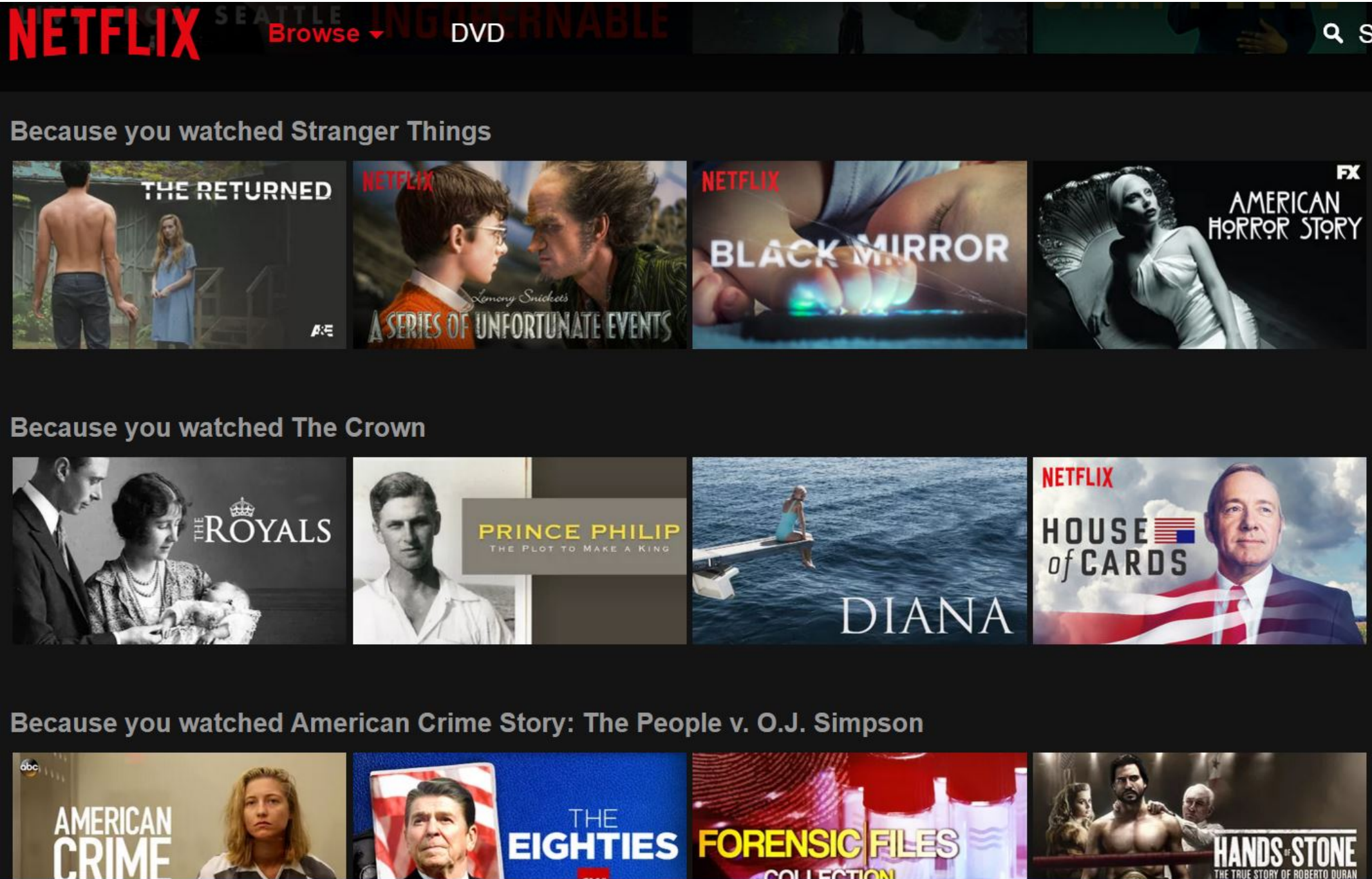
Specific, measurable questions

Do Tiktok videos with dogs go more viral?

- Let's make it better!
- Do tiktok viewers between the ages of 18 and 20 view more videos with a dog in at least 50% of the frames than videos with cats in 50% of frames within the first 10 days of being posted?
 - Topic
 - Features
 - Outcome
 - Timeframe
 - Comparison group

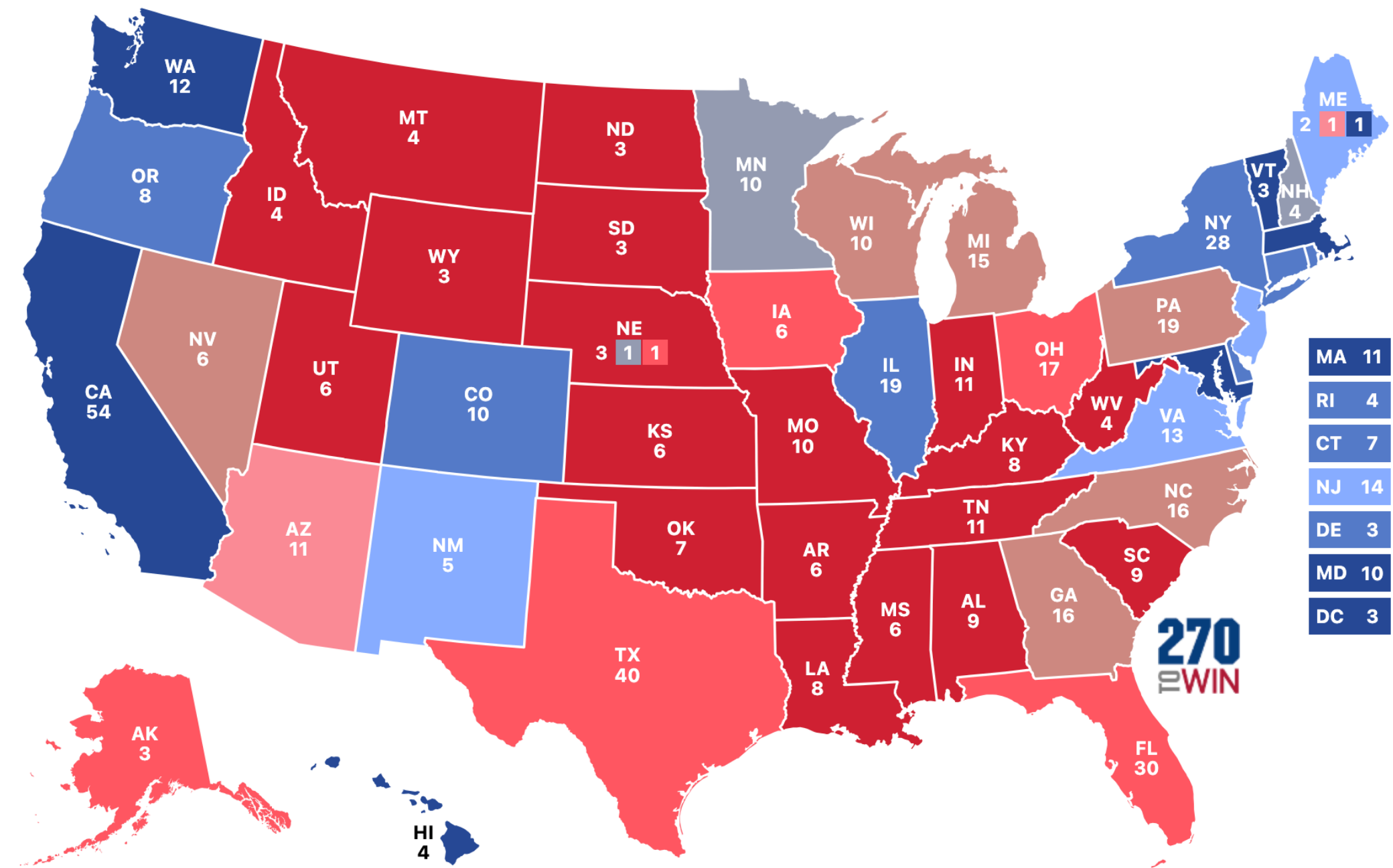
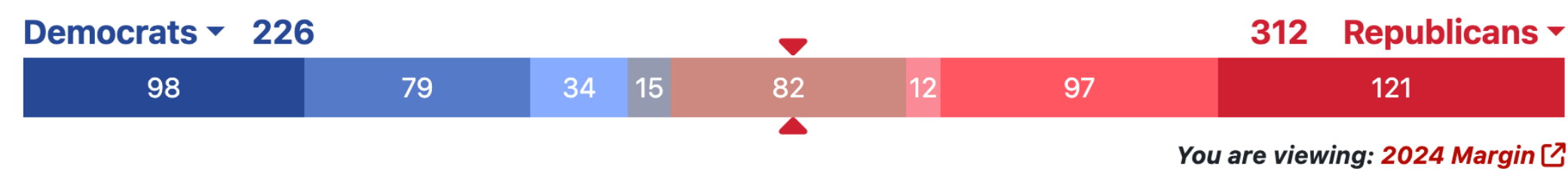
Data science questions can have different goals

Describe or understand patterns in the data



Do people who watch Stranger Things also watch Game of Thrones?

Predict what will happen in the future



Will the Republican or Democratic party win the presidential election in 2028?

Data science questions can have different goals

Test cause-and-effect between variables

Only 1 left in stock - order soon.

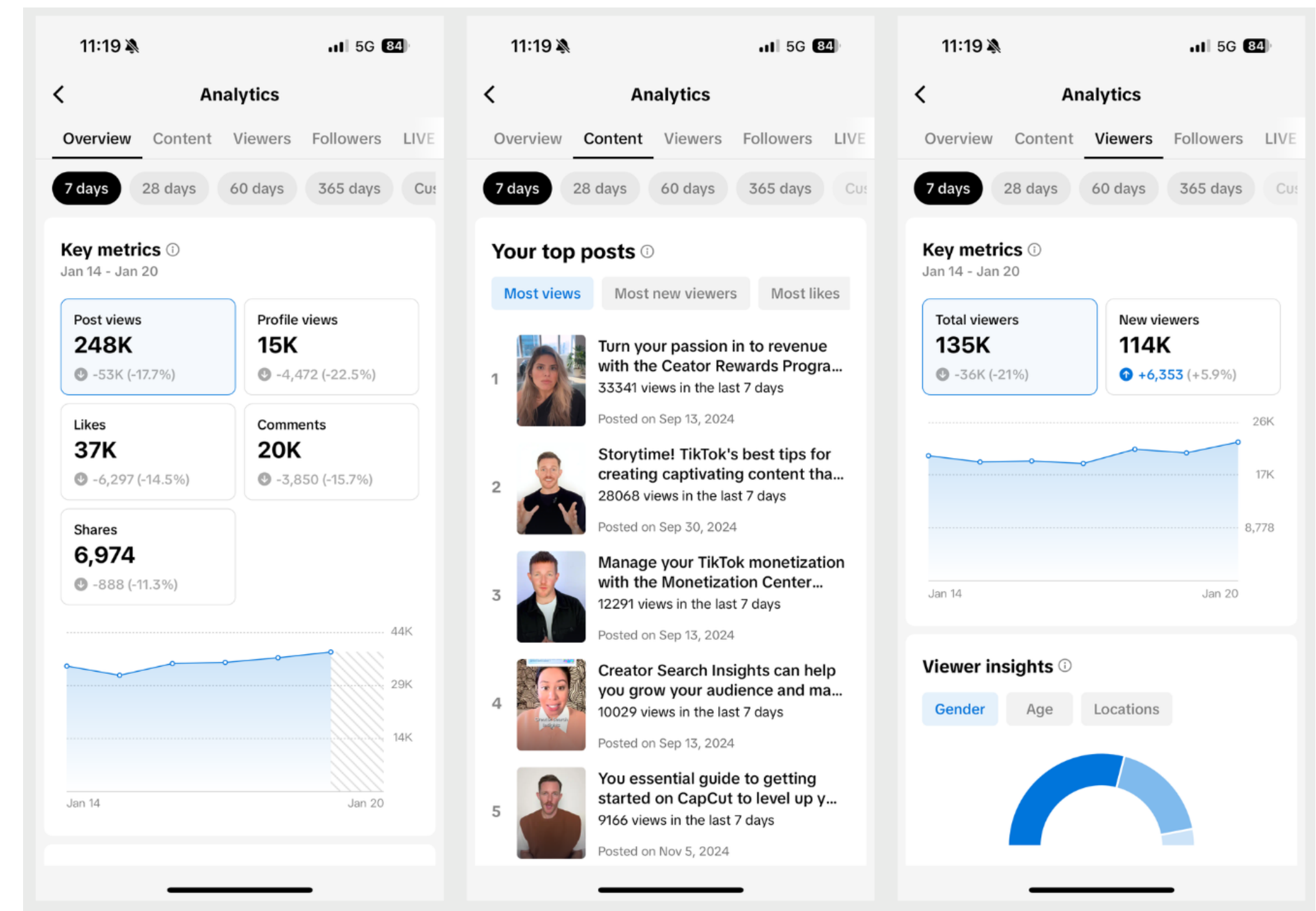
Add to Cart

Buy Now

 Secure transaction

Does adding a “Buy Now” button increase sales over a 1-year period?

Decide what to do or optimize something to get the best outcome



At what time of day should I post my TikTok videos to get the most views?

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Which of the following is a data science question aimed at testing *cause-and-effect*?

- (A) What percentage of customers prefer online shopping?
- (B) Does offering free shipping increase the average number of orders per customer?
- (C) How many visitors came to the website last month?
- (D) When should I offer discounts to increase customer purchases?
- (E) I'm not sure / I'm confused

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Which of the following is a data science question aimed at *describing*?

- (A) What will UCSD's acceptance rate be next year?
- (B) What percentage of UCSD students went to Taco Bell at least once this week?
- (C) If UCSD extends library hours during finals, will average study time increase?
- (D) When should I host my office hours so that most students will come?
- (E) I'm not sure / I'm confused

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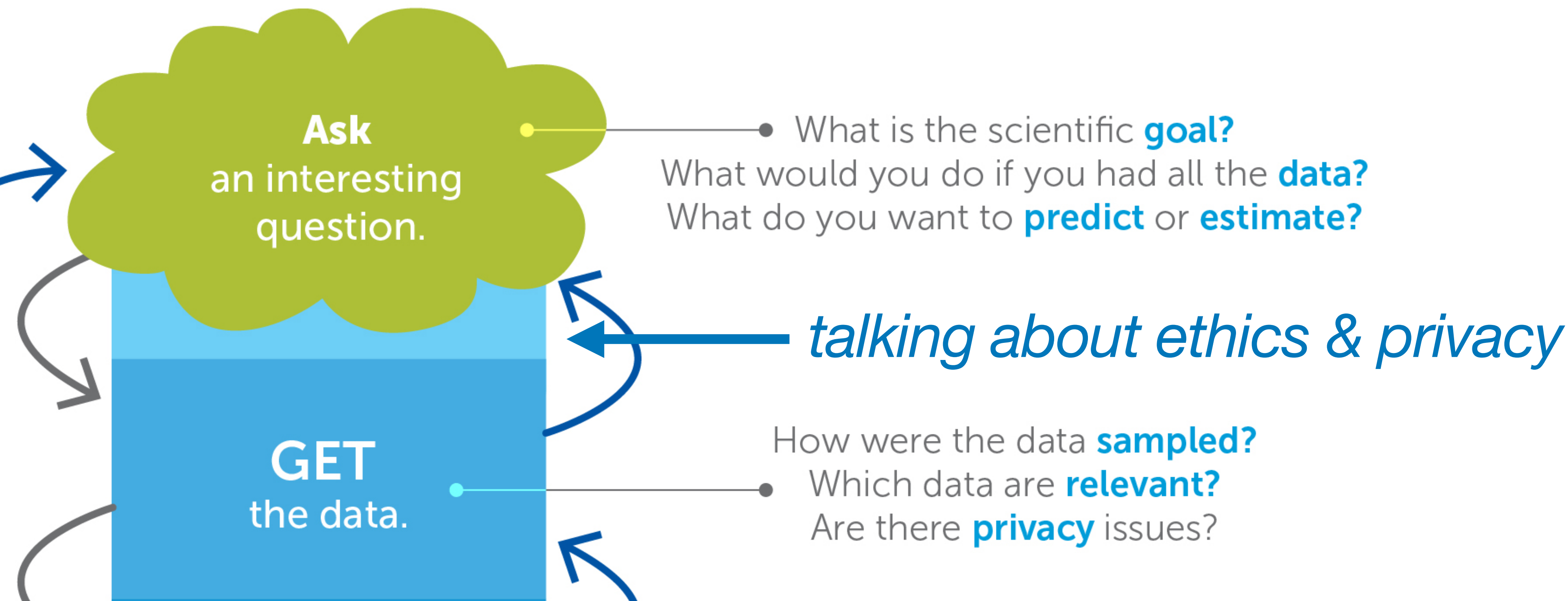
Which of the following is a data science question aimed at *describing*?

- (A) What will UCSD's acceptance rate be next year? (Predict)
- (B) What percentage of UCSD students went to Taco Bell at least once this week? (Describe)
- (C) If UCSD extends library hours during finals, will average study time increase? (Cause-and-effect)
- (D) When should I host my office hours so that most students will come? (Decide/optimize)

Ethics and Privacy

Why talk about ethics & privacy upfront?

- The choices you make early on (what question to ask, what data to collect), can create harm, bias, or privacy risks. Talking about it up front helps:
 - Avoid preventable harm
 - Build trust with users
 - Save time (much harder to fix ethics/privacy issues *after* the data is collected)



Reading(s) 2: *Ethics*

focuses on
responsibility and practice

Data's day of reckoning

We can build a future we want to live in, or we can build a nightmare. The choice is up to us.



By [DJ Patil](#), [Hilary Mason](#) and [Mike Loukides](#)

July 31, 2018 • 18 minute read

DOI:10.1145/1743546.1743558

Arvind Narayanan and Vitaly Shmatikov

Privacy and Security Myths and Fallacies of “Personally Identifiable Information”

Developing effective privacy protection technologies is a critical challenge for security and privacy research as the amount and variety of data collected about individuals increase exponentially.

THE DIGITAL ECONOMY relies on the collection of personal data on an ever-increasing scale. Information about our searches, browsing history, social relationships, medical history, and so forth is collected and shared with advertisers, researchers, and government agencies. This raises a number of interesting privacy issues. In today's data protection practices, both in the U.S. and internationally, “personally identifiable information” (PII)—or

Any information that distinguishes one person from another can be used for re-identifying data.

are commonly used for authenticating an individual, as opposed to those that violate privacy, that is, reveal some sensitive information about an individual. This crucial distinction is often overlooked by designers of privacy protection technologies.

The second legal context in which the term “personally identifiable information” appears is privacy law. In the U.S., the Privacy Act of 1974 regulates the collection of personal information by government agencies. There is no

Theme:
why ethics and privacy
can't just be an
afterthought in our
practice of data science

why de-identification fails
and what privacy
protection should look like

Reading(s) 2: *Ethics*

- Readings can be dense and overwhelming—part of this course is learning how to read academic papers!
- Use the Reading Guide to focus on the key ideas
- Reading 2 Quiz: **due Wednesday of Week 3!**
 - 10 MC questions
 - 60 mins to complete it
 - You may use the reading & reading guide!

R2: Reading Guide

1. [Loukides M, Mason, H & Patil DJ, "Data's Day of Reckoning"](#)
2. [Narayanan and Shmatikov, "Myths & Fallacies of Personal Identifiable Information"](#)

Why we're reading these

These two readings explain why ethics and privacy can't just be an afterthought in our practice of data science. The first focuses on responsibility and practice, and the other on why de-identification fails and what privacy protection should look like.

Key ideas (and why they matter)

We may ask you about these key ideas in exam 1.

"Data's Day of Reckoning"

1. Data-driven products can be weaponized (misuse, abuse) when incentives don't align with the public good.
2. Ethics must be integrated into everyday practice and data science education, and not treated as an optional afterthought.
3. Use concrete mechanisms like checklists and "stop-the-line" culture to make

Learning Goals

1. Define **ethics** and **privacy** and explain why they are important
2. Examine how ethical lapses and privacy failures can harm people at every stage of the data science process
3. Identify practical steps that data scientists can take **address ethical concerns** and **maximize privacy**
4. A code of conduct for **responsible data science**

Some definitions

- **Ethics:** Principles and standards that guide choices about how data is collected and how models are built and used
 - *What is right, fair, and responsible in a given situation?*
- **Privacy:** A person's ability to control access to themselves and their own information
 - *What can others observe, collect, use, or share about you?*



Unintended harm



- **The Cobra Effect:** the unintended consequences that result from an attempt to solve a problem. Also known as a **perverse incentive**
- **Takeaway:** think about all the potential consequences of your data science goal *before* pursuing it

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I believe that most of the data
collected about me stays private

- (A) strongly disagree
- (B) disagree
- (C) neither agree nor disagree
- (D) agree
- (E) strongly agree

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course code:
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Sign in with your UCSD email!

I don't worry about privacy issues
because I have nothing to hide

- (A) strongly disagree
- (B) disagree
- (C) neither agree nor disagree
- (D) agree
- (E) strongly agree

Unintended harms: data collection

Should we collect this?

Facebook–Cambridge Analytica scandal (2014–2018)

- Personal data from millions of Facebook users (+ their friends) were collected (without consent) by British consulting firm *Cambridge Analytica*
- Used this data to target tailored political ads to specific people

Revealed: 50 million Facebook profiles harvested for Cambridge Analytica in major data breach

FTC Imposes \$5 Billion Penalty and Sweeping New Privacy Restrictions on Facebook

FTC settlement imposes historic penalty, and significant requirements to boost accountability and transparency



Unintended harms: data sharing

Should we share this?

Strava heatmap (2018)

- GPS data “accidentally” revealed sensitive locations and movement patterns around military bases
- Default data sharing setting set to public

Fitness tracking app Strava gives away location of secret US army bases

Data about exercise routes shared online by soldiers can be used to pinpoint overseas facilities

- Latest: Strava suggests military users ‘opt out’ of heatmap as row deepens



📷 A military base in Helmand Province, Afghanistan with route taken by joggers highlighted by Strava. Photograph: Strava Heatmap

Unintended harms: data analysis

Should we analyze this?

Target pregnancy prediction (2012)

- Can predict when someone is pregnant (+ their due date!) by analyzing their purchasing patterns
- Used information for timed, targeted ads or coupons on “pregnant people essentials” like supplements (calcium, magnesium, zinc), unscented lotion, etc.

TECH

How Target Figured Out A Teen Girl Was Pregnant Before Her Father Did

By [Kashmir Hill](#), Former Staff. Welcome to The Not-So Private Parts where technology & privacy collide

Published Feb 16, 2012, 11:02am EST, Updated Aug 11, 2022, 04:17am EDT



Unintended harms: data modeling

Should we model or predict this?

Amazon recruiting model (2018)

- Tried to use a machine learning model to predict who to hire
- Model was “taught” or trained with historical hired resumes (mostly men), which led the algorithm to discriminate against women
 - This is called a *class imbalance problem*: when the data itself are biased

Amazon ditched AI recruiting tool that favored men for technical jobs

Specialists had been building computer programs since 2014 to review résumés in an effort to automate the search process

NEWS & COMMENTARY

Why Amazon's Automated Hiring Tool Discriminated Against Women

Unintended harms: data scraping

Should we build models with this?

LLMs

- Social media
- Google books
- Art, images

 ENGLISH WHO WE ARE WHAT WE DO



[< NEWS](#)  

28 May 2026 Also available in Español, Français, العربية

Global: Enormous data pipelines powering major generative AI systems are rooted in mass invasions of privacy by design

Events Alumni CALIFORNIA LAW REVIEW   

Volume 113 • October 2025 • Daniel J. Solove • Woodrow Hartzog • Article

The Great Scrape: The Clash Between Scraping and Privacy

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course code:
WSTAPX

Sign in with your UCSD email!

Companies should be allowed to use my data if they give me more personalized recommendations or results

- (A) strongly disagree
- (B) disagree
- (C) neither agree nor disagree
- (D) agree
- (E) strongly agree

Think of an example where this is true in an app or product that you use (~1min)

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course code:
WSTAPX

Sign in with your UCSD email!

It's my responsibility (not companies)
to manage my privacy settings

- (A) strongly disagree
- (B) disagree
- (C) neither agree nor disagree
- (D) agree
- (E) strongly agree

Elaborate to a neighbor! (~3mins)